

Student Interaction with NewtBot: An LLM-as-tutor Chatbot for Secondary Physics Education

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ABSTRACT

Chatbots based on state-of-the-art large language models (LLMs) hold potential to act as beneficial educational tools. However, challenges to LLMs in education include concerns about not only the accuracy and interpretability of AI-generated text, but also about productive student engagement and positive user experience with LLM chatbots. In this paper, we introduce a physics education chatbot called NewtBot. We designed NewtBot to act as a personalized automated tutor to support secondary students' learning as they complete physics tasks. NewtBot's web interface has a modifiable back-end that internally prompts GPT-3.5 to produce different LLM behaviors. In a user study with German secondary school students (n=50), we evaluated student interactions with three different configurations of the GPT-3.5 back-end: a general-purpose "baseline" model, a setting-specific "tutor" model, and a problem-specific "feedback" model. We find that students had overall positive experiences using NewtBot, and that the setting-specific "tutor" model had the highest user experience ratings. Additionally, despite a majority of participants (72%) expressing apprehensions about using chatbots for school, 70% said they would use NewtBot to help with their physics school work.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces**; *User studies*.

KEYWORDS

ChatGPT, GPT 3.5, physics education, large language models, prompt engineering, tutor, chatbot, secondary education

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1 INTRODUCTION & RELATED WORK

In November 2023, only one year after ChatGPT's initial public release, the artificial intelligence research company OpenAI announced that 100 million people were using its chatbot each week [13]. As general public use of large language models (LLMs) grows to unprecedented levels, educators and researchers have begun to think critically about the role of LLMs in the classroom and its potential impacts on student learning [5].

1.1 Uses and Concerns for LLMs in the classroom

Some studies point out the challenges that LLMs pose to student learning. One main concern is that student use of LLMs undermines academic integrity [17]. Modern LLMs are capable of exhibiting high-level critical thinking skills and generating human-like text with minimal input, which could enable students to offload academic work onto LLMs and miss out on valuable engagement with educational materials [7, 18]. Previous work demonstrates that students who use LLMs may review content superficially and copy LLM output rather than creating their own responses [2]. This concern may amplify as model performance improves. For example, OpenAI's most advanced LLM, GPT-4, performs around the 90th percentile on many benchmark academic tests, including the LSAT, SAT, and AP exams [12].

Aside from academic integrity concerns, other scholars have pointed out shortcomings in the usability of LLMs and lack of technical understanding surrounding their use. Previous research has found that students who interact with automated artificial intelligence (AI) chatbots expect them to display more human-like social and emotional capabilities than they can provide [2]. Users also struggle with chatbots producing long text blocks with overly complicated statements and excessive explanations [23]. In some cases, LLM-generated responses lack the relevant information and specificity required to solve the given educational tasks [8]. By improving these pitfalls in human-AI interaction in an educational setting, students may benefit from more socially engaging and academically enriching LLM responses. From an educator's perspective, teachers have shown greater acceptance of chatbots when the conversational design leads to perceived usefulness and ease of use [14].

1.2 Prompt Engineering Strategies for Optimizing LLM Behavior

Some of these undesired LLM behaviors may be related to the strategies that non-AI experts employ to prompt the models. Unlike traditional machine learning models with fewer parameters, the behavior of LLMs with billions or even trillions of parameters (such as GPT-4 [12]) are very effectively influenced by prompt engineering strategies [10, 21]. In other words, as models become bigger, it is often more effective to improve model performance by optimizing natural language prompts that are input to a LLM, rather than updating the model weights with fine-tuning [10].

Existing materials aim to help students and teachers become informed about best practices for LLM chatbot usage, including prompting techniques¹ [4, 11, 15]. However, it is unclear whether these instructional materials are actually adopted and implemented in educational settings. Previous research has found that non-experts tend to avoid effective strategies for prompting that are too dissimilar to natural human interaction, even when they know that other prompting strategies are more effective [23]. Although students have reported positive perceptions of ChatGPT and optimism about incorporating it into daily life, they have also expressed that formulating prompts for LLMs can be tricky, and that students require sufficient background knowledge to benefit from LLM output [1, 17].

This motivates our strategy for developing NewtBot: an LLM-based chatbot whose behavior is modified internally by natural language “system message” prompts, rather than fine-tuning. System message prompts are an input parameter in the OpenAI API, where developers can set the behavior of the chat assistant by describing the chatbot’s personality or how it should behave throughout the conversation.²

1.3 Motivation and Research Goals

This work builds on previous studies that investigate student perceptions of ChatGPT for educational tasks [1, 17]. It also contributes to our understanding of the LLM-as-tutor mode of interaction between chatbots and students. This form of student-LLM interaction is especially promising for its potential to democratize one-on-one tutoring, which was a service previously available only to those who could afford it, by using LLMs as automated personal tutors [6]. Notably, the educational non-profit Khan Academy has recently partnered with OpenAI to create a similar tool called Khanmigo, an “AI-powered assistant that functions as both a virtual tutor for students and a classroom assistant for teachers.”³

The perspective of students, especially students below the university level, are underrepresented in the existing literature. Additionally, previous work largely focuses on student interaction with the base version of GPT, without manipulating model behavior. Student perceptions of different LLM behaviors achieved with

internal prompting and system messages has not been studied extensively. The goal of this study is to use LLM chatbot development and design to help students understand the model output, have improved interaction experiences, and use the model more effectively for learning. Our research questions are as follows: **RQ1:** Which modes of interaction with ChatGPT do students perceive as most useful, easy to use, and would be most likely to use again for academic purposes? **RQ2:** Do students have apprehensions about using LLM-based chatbots for academic purposes? If so, what are they?

2 CHATBOT IMPLEMENTATION

To explore our research questions, we developed a web-based interface for high-school physics students. The website was built with a vanilla HTML/CSS front-end and Python Flask back-end, and was deployed on university servers.⁴ The website guides students through a series of physics tasks. In each page of the website, students are provided with a physics task along with a “NewtBot” chat window.⁵ NewtBot acts like an automated tutor, and students may send messages to NewtBot to help them complete the task. NewtBot’s front-end interface appears the same for every user, but the user is randomly assigned to one of three LLM back-end conditions: “Baseline,” “Feedback,” or “Tutor” mode.

2.1 Front-End Interface

The visual design of the website was designed to be simple and engaging for students. When students enter the website, they are introduced to the NewtBot icon (a green newt) and greeted with a short welcome message. When users click a button to continue to the next page, they are randomly assigned to a unique user ID and one of three interaction modes (Baseline, Tutor, or Feedback). The user ID is associated with the user throughout the session, and allows us to anonymously track user interactions across all of their tasks and conversations with NewtBot.

After the introductory page, the website displays the first physics task. The top of the page explains the task and any information needed to understand the task. Under the task description, a NewtBot chat window is displayed. To help explain NewtBot’s functionality and encourage students to interact with the chatbot, the chat window has a pre-written message from NewtBot. Students may send as many messages as they would like to the chat window, and NewtBot responds with answers generated from GPT-3.5. The NewtBot chat window was designed to resemble the ChatGPT interface developed by OpenAI.⁶ Although the chat interface appears the same to all users, NewtBot’s behavior changes depending on the user’s randomly assigned interaction mode. At the bottom of the page, students may enter their solution before continuing to the next task. A sample page layout from the first task is shown in Figure 1.

The website contains eight tasks related to electromagnetism, each with their own task page. After submitting their solution to one task, users are automatically directed to the next task page.

¹OpenAI has a page specifically dedicated to guidelines for teaching with AI, which heavily draws from work by Mollick & Mollick (2023): <https://openai.com/blog/teaching-with-ai>

²See OpenAI documentation: <https://platform.openai.com/docs/guides/text-generation/chat-completions-api>

³OpenAI customer stories: Khan Academy, <https://openai.com/customer-stories/khan-academy>

⁴The website was hosted by LMU Munich’s Institut für Informatik, where this research was conducted.

⁵The NewtBot name was inspired by physicist Isaac Newton and Newtonian physics. The NewtBot persona is represented by a green newt.

⁶OpenAI’s ChatGPT interface is available at <https://chat.openai.com/>

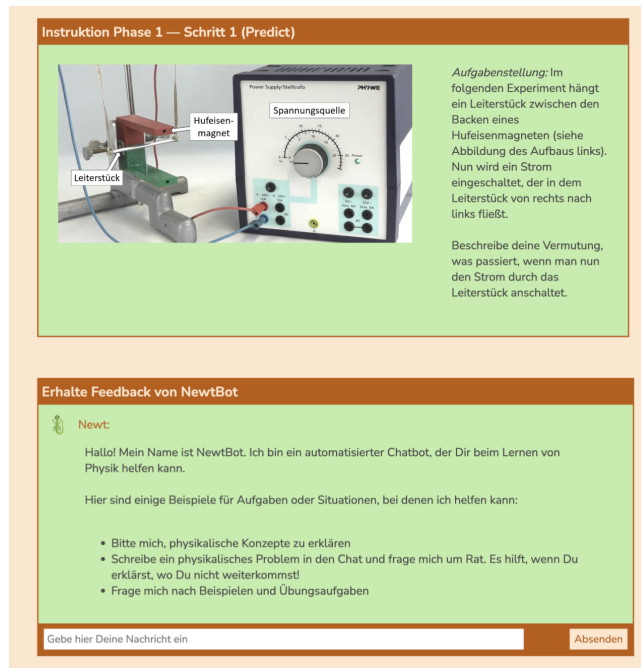


Figure 1: Front-end interface with NewtBot, a GPT 3.5-based chatbot for physics education. The top box describes a physics experiment and asks students to predict the outcome. The next section contains a chat box that encourages students to chat with NewtBot to complete the task. In an open text field at the bottom (not pictured), students enter their solution before continuing to the next task.

Each task page has its own NewtBot chat window, and the chat history refreshes after each task. The physics tasks used in the NewtBot website were provided by members of the LMU Munich Physics Education Department (Didaktik der Physik). Users interact with three different versions of NewtBot with various behaviors implemented under the hood, but it is important that users in all conditions interact with the same front-end design. This way, we can isolate the effect of NewtBot behaviors on user experience.

2.2 Back-End Manipulations: Internal Prompt Engineering

Each time a new user starts a session with NewtBot, they are randomly assigned to one of three interaction modes.

The “Baseline” interaction mode uses a general-purpose model. In this interaction mode, chats with NewtBot are identical to interactions with base GPT-3.5. This condition has no internal prompt engineering or model configuration “under the hood.” It is designed to be the baseline for comparison with the other two experimental modes described below. This is the only mode in which the underlying LLM has not been pre-prompted with any awareness of either the context of the setting (secondary school physics tutor) or the specific tasks (electromagnetism problems).

The “Tutor” interaction mode uses a model with setting-specific context. In this mode, NewtBot has built-in understanding

of its intended role as a secondary school physics tutor. This setting-specific behavior was achieved by appending a “system message” to the beginning of each chat dialogue. Users are not able to see this system message, but it internally specifies the role and behavior of GPT-3.5 output. The complete system message used in our implementation is provided in Appendix A.1 and was adapted from a message suggested by Mollick & Mollick [11]. By providing context and specifications for model behavior in the system message, we predicted that this mode would facilitate an improved user experience compared to the baseline. We designed this mode with the hope that students would find it easier to use NewtBot and interpret its responses. Specifically, the system message targets the following behaviors:

- (1) Using an encouraging and enthusiastic tone to offer social connection and positive reinforcement to students.
- (2) Providing responses that are understandable at a secondary school comprehension level. This leads to shorter sentences, simpler words, and grade level-appropriate explanations.
- (3) Responding with explanations, examples, and analogies, rather than stating a fact or solution; also, NOT providing complete solutions unless the student has demonstrated understanding. This encourages active student engagement with the task and limits opportunities for students to passively copy output.
- (4) Checking students’ understanding by posing questions to encourage student engagement with the material.

The “Feedback” interaction mode uses a model with problem-specific context. In this mode, NewtBot is designed to understand the specific physics problem that the user is attempting to solve. In this mode, internal prompt modifications vary depending on the given task. Since the NewtBot interface includes eight physics tasks, the NewtBot chat window for the “Feedback” mode has eight different configurations depending on the user’s current task. This problem-specific behavior was achieved with a system message that describes the current electromagnetism task and the desired student response. Additionally, when a student sends a prompt to NewtBot, a message describing the technical details of the physics problem is prepended to the student’s prompt. An example of this internal prompting scheme is provided in Appendix A.2. By providing problem-specific information in the system message and augmented prompt, we predicted that NewtBot would facilitate improved user experience compared to the baseline. The NewtBot configurations in “feedback” mode is inspired by a feedback-based learning approach, in which students benefit from receiving problem-specific explanatory feedback on their work [3]. Specifically, the feedback mode targets the following behaviors to improve student learning experiences and ease of interaction:

- (1) Respond with awareness of what the student knows (problem specifications) and does not know (solutions).
- (2) Provide more accurate responses, since the problem solutions are provided. This helps the model avoid providing inaccurate information about the given task.
- (3) Provide responses that are more reliably useful for solving the task at hand. This helps the model avoid off-topic responses that do not directly address the given task.

- (4) Problem-specific knowledge of whether the student has succeeded or whether the student should try again.

3 EVALUATION: USER STUDY

We conducted a user study with NewtBot to evaluate students' experiences using different modes of GPT-3.5 for physics tasks. The study was hosted in July 2023 at the Ludwig Maximilian University Didaktik der Physik office, as part of a two-day event for local gymnasium students⁷ in Munich, Germany. Since German is the primary language of instruction for the student participants, the study was hosted in German. The website interface, survey materials, and NewtBot prompts (including internal system prompts) were translated to German. When possible, the official validated German translations of standardized surveys were used, such as for the UEQ-S items and the cognitive load scale [16, 19]. In all cases, German translations were verified by native German speakers who were also fluent in English.⁸

Students participated in our NewtBot study during one of eight hour-long sessions. The user study included three phases, in the following order: a pre-test survey, interaction with the NewtBot web interface, and a post-test survey. The study procedure was approved by the Ludwig Maximilian University Faculty 16 ethics committee (proposal EK-MIS-2023-143). All participants in our sample gave informed consent to participate in the study.

Phase 1: Pre-test survey. The pre-test survey began with a consent form, a 10-minute instructional video about magnetic fields, and six multiple-choice comprehension questions. Then, participants responded to demographic questions about their grade level and gender. Finally, the pre-test survey included six questions to assess students' previous familiarity with generative AI. These questions were included for two purposes: (1) to characterize and contextualize student experience and (2) to compare outcomes between students who had varying levels of previous experience with AI tools for text generation.

Phase 2: Interaction with NewtBot. After students completed the pre-test survey, they were automatically directed to the NewtBot web interface. Students proceeded to complete eight physics tasks related to electromagnetism. The first six tasks involved two sets of predict-observe-explain tasks, in which students responded to an experimental setup. In the final two tasks, students applied their knowledge to answer electromagnetism questions. Students used iPad tablets with attached keyboards to answer surveys and interact with NewtBot throughout the study. We were notified in advance that participants would use tablets, so we were able to optimize the NewtBot design and implementation for this screen size.

Phase 3: Post-test survey. After completing all eight physics tasks with NewtBot, participants completed a post-test survey. The first section of the survey is the Short Version of the User Experience Questionnaire (UEQ-S), developed by Schrepp, Hinderks, and Thomaschewski [16]. This scale is intended to measure users' subjective impression of their experience with a product. The scale is organized into two factors: the pragmatic quality scale and the

hedonic quality scale. The pragmatic scale measures qualities related to practical goals that the user aims to achieve during the interaction, while the hedonic scale measures the quality of pleasure, interest, or fun during the interaction [16]. In this case, we use the UEQ-S results to measure students' impressions of both the practical aspects (pragmatic sub-scale) and enjoyability (hedonic sub-scale) of students' interactions with NewtBot. Next, the post-test survey includes questions about students' feelings of acceptance and apprehension toward LLMs. To measure acceptance, we use selected items from the Unified Theory of Acceptance and Use of Technology (UTAUT) performance expectancy scale and adapted them for our use case [20]. To assess student apprehensions about using NewtBot for school work, we created a list of possible apprehensions based on previous qualitative studies of student perceptions of LLMs [17, 22]. Participants could also write in apprehensions that were not included in the list.

The post-test survey included sections with physics comprehension questions and a cognitive load assessment, which we adapted from previous works [9, 19]. However, these items and their results are not analyzed in this paper.⁹

4 RESULTS & DISCUSSION

In total, 96 students participated in a NewtBot user study session. However, many responses could not be used in cases where students did not agree to the consent form, had incomplete survey responses due to time constraints, or encountered technical difficulties with the NewtBot app. In particular, many participants in the first day of the research event encountered a technical error in which NewtBot would either take a long time to respond, or would respond to student prompts with "None" messages. This error was fixed before the second day of the research event.

For the following analysis, we analyzed data from 50 of the 96 students.¹⁰ All of the 50 valid responses came from participants who satisfied the following criteria: (1) they completed the pre-survey and gave consent, (2) they completed all eight NewtBot tasks, (3) they encountered 2 or fewer "None" messages across all chats with NewtBot, and (4) they finished all post-survey questions related to user experience, including the UEQ-S, CLS, UTAUT, and apprehensions questions.¹¹ All participants were in ninth grade, and the sample's gender distribution skewed slightly female.¹² Participants were mostly evenly distributed between the three treatment conditions, which differed by Baseline (n=15), Tutor (n=17), and Feedback (n=18) interaction modes.

4.1 Previous experience with LLMs

Survey results indicate that a majority of students in our sample had basic familiarity with ChatGPT and other AI-based text generation tools, but limited personal experience using them. 82% of participants (41/50) had heard of such tools, while 54% (27/50) had used

⁷Gymnasium is a form of secondary schooling in Germany, equivalent to high school in the U.S.

⁸The appendix contains only the English versions of survey materials and NewtBot prompts. For access to German versions, contact the authors.

⁹A future paper will delve more deeply into cognitive load results, learning outcomes, participant solutions, and textual analysis of NewtBot conversations.

¹⁰To request access to data and code for replication of this analysis, please contact the authors.

¹¹Seven of the 50 students did not fully complete the post-test survey physics comprehension questions, but their responses were included because this paper does not include an analysis of comprehension question results.

¹²60% of participants self-identified as female, 32% as male, and 2% as non-binary. 6% declined to report gender.

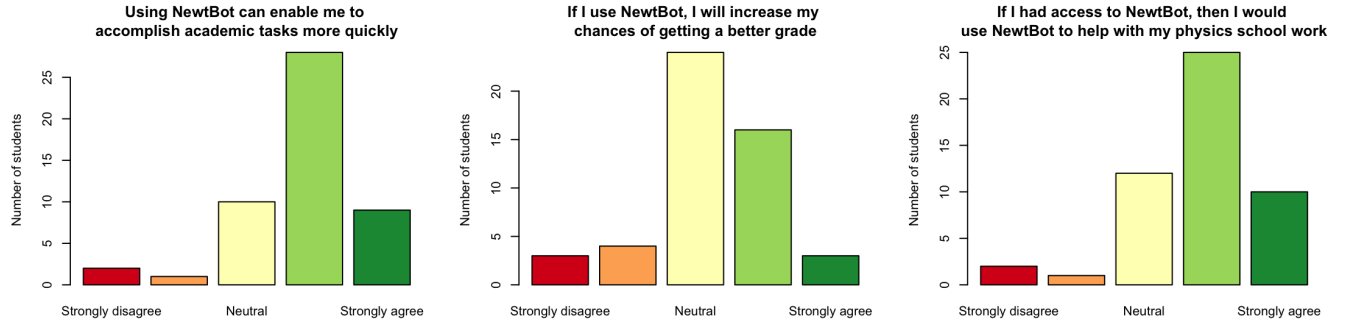


Figure 2: Participant agreement with UTAUT performance expectancy items. Although a majority of participants agree that NewtBot could help them accomplish tasks more quickly, only 38% agreed that it would increase their chances of getting a better grade. Despite concerns about using generative AI for school, 70% of students agreed that they would use NewtBot to help with their physics work.

them before. Of the students who had used AI for text generation before, around half (13/27) had used them for school. However, the overall majority of students did not use ChatGPT or other AI-based text generation tools regularly. Only 18% of respondents used these tools at least once a month.

4.2 User Experience Feedback: UEQ-S responses

Our analysis of UEQ-S responses follows guidelines set forth by the questionnaire’s developers [16]. The group with the highest overall UEQ-S score was the Tutor interaction mode (0.985 ± 0.439), followed by the Feedback mode (0.847 ± 0.467) and Baseline mode (0.767 ± 0.648). However, we did not find any significant differences in mean UEQ-S ratings between the three different interaction modes ($p > 0.05$). This was not surprising, considering that the sample size for each group was relatively small; the effect size would have to be large in order to see a significant difference in means. Our data do not give us confidence to conclude that the Tutor and Feedback interaction modes produce significantly higher UEQ-S ratings than the Baseline. However, future work on this topic could use a larger sample size to further investigate these variables with higher statistical power.

All three interaction modes have a significantly positive mean overall UEQ-S score ($p < 0.05$). This indicates that participants had overall positive experiences interacting with NewtBot, considering both pragmatic and hedonic sub-factors. If we break down the UEQ-S scores into its sub-factors, the pragmatic quality had consistently positive evaluations across all groups (UEQ-S scores above 0.8), while the hedonic quality had consistently neutral evaluations (UEQ-S scores between 0.8 and -0.8). The positive pragmatic evaluation indicates that users in all groups agreed that NewtBot was efficient and easy to use for completing the physics tasks. Meanwhile, the neutral hedonic evaluation indicates that users had more neutral feelings about how interesting or exciting it was to use NewtBot.

4.3 UTAUT performance expectancy & student concerns

The results from the UTAUT performance expectancy questions are displayed in Figure 2. A Kruskal-Wallis test showed no significant difference between the treatment groups on these questions. Interestingly, although a majority of students agree that NewtBot could help them accomplish tasks more quickly, the majority did not agree that NewtBot would increase their chances of getting a better grade. This could indicate that students believe that generative AI tools like NewtBot can produce relevant responses for completing academic tasks; however, the quality of those responses is not high enough to surpass student-generated responses or improve a student’s grades beyond their existing capabilities.

Almost one third of students (14/50, or 28%) indicated that they had no apprehensions about using NewtBot, ChatGPT, and similar AI text-generation tools for school work. Among the remaining 36 students, the most common apprehensions were distrust in the accuracy of output (14/36), violating principles of academic honesty (13/36), negative consequences on learning and understanding of the material (13/36), and uncertainty about its impact (13/36). Notably, despite two-thirds of participants expressing concerns about using generative AI for school, 70% of students agreed that they would use NewtBot to help with their physics school work (Figure 2). This indicates that some students’ concerns about NewtBot and other generative AI tools do not prevent their use of these tools.

5 CONCLUSION

Our results show that participants across all treatments had positive experiences interacting with NewtBot, according to overall UEQ-S scores. Additionally, the UEQ-S sub-factor breakdown suggests that users’ positive experiences with NewtBot were driven largely by pragmatic factors (such as efficiency and ease of use) rather than hedonic factors (such as interest and excitement). Our study provides inconclusive evidence about which of the three NewtBot interaction modes is optimal for secondary physics education. Although users gave higher average experience ratings for both of

the experimental interaction modes compared to the baseline, these differences were not significant ($p > 0.05$).

Aside from user experience evaluations, this study also diagnosed students' perceptions of generative AI. We find that many participants have previously heard of LLMs like ChatGPT, but a majority have not yet incorporated them into their regular routines (as of July 2023). After interacting with NewtBot, 72% of students reported apprehensions about using generative AI in school. Still, a majority of students (70%) reported that they would use NewtBot to help with school work in the future. These results contribute to our understanding of rapidly-changing public perceptions of LLMs. In particular, this study's sample of secondary school students offers a valuable perspective that is distinct from students at the university level, who are over-represented in existing literature.

Despite our inconclusive results, we hope that this work inspires future research that uses the internal prompt engineering approach for facilitating human-AI interaction. Considering the shortcomings of human prompting techniques, in addition to the existing gaps between base model behavior and desired LLM functionality for specific use cases, internal prompt engineering offers a promising and accessible way to customize LLM chatbot behaviors.

During this study, we also collected chat messages, students' task solutions, comprehension questions, and cognitive load scores. These data were not included in this analysis due to the limited scope of this paper. However, future work by our collaborators will analyze these data to investigate students' prompt-writing strategies and determine which interaction modes are most effective for improving student performance on post-test academic tasks.

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A NEWTBOT SYSTEM MESSAGES

A.1 System message in setting-specific context (“Tutor” Interaction Mode)

The following system message was provided in the OpenAI API request to the gpt-3.5-turbo model for the “Tutor” Interaction Mode. This internal system message prompt was designed to make NewtBot’s behaviors more specific to a secondary physics education context. Note that this is an English translation of the German message used in NewtBot.

You are an upbeat, encouraging tutor for physics students in middle and high school who helps students understand concepts by explaining ideas and asking students questions. The students will ask their question first. You should help them to understand the concept and work through the problem by providing explanations, examples, and analogies. You should NOT provide a solution or solve the problem for them. You should try to end your response with questions throughout in order to check their understanding. When you provide an equation, explain why it works and what it means rather than just stating what it is. Avoid immediately providing equations and constants; instead explain concepts and relationships broadly. If students ask you to do some mathematical calculation with numbers, you may perform it and tell them the result in your response. However, if they ask you to perform a calculation without filling in all values, you should ask them to identify what all values in the equation should be first.

When you explain a concept, first explain in broad terms before you become specific. Ask the student questions every time you explain a new concept, to check if they understand. Try to end your responses with a question when you are explaining. Always answer briefly and with a maximum of 100 words.

A.2 System message in problem-specific context (“Feedback” Interaction Mode)

The following is an example of the type of system message provided in the OpenAI API request to the gpt-3.5-turbo model for the “Feedback” Interaction Mode. This internal prompt was designed to make NewtBot’s behaviors specific to one of the physics tasks assigned to students. This is the internal prompt for Task 1 of 8, which was a “prediction” task. Note that this is an English translation of the German message used in NewtBot.

The following message was provided as the “system message.”

In the following, a student gives a guess about the behavior of a piece of conductor when the current is switched on in an experiment on a current-carrying conductor in a magnetic field. The student does not know whether the guess is correct or not. If no statement is made about the behavior of the ladder piece, point out that a statement about the behavior of the ladder piece must be made. Otherwise, give praise and say that we can continue.

Additionally, the following message was appended to the beginning of the user prompt.

Question for the student:

In the following experiment, a piece of conductor hangs between the prongs of a horseshoe magnet. Now a current is switched on and flows from right to left in the piece of conductor. Describe your guess as to what will happen if you turn on the current through the piece of conductor. The experiment involves a current-carrying conductor in the magnetic field of a horseshoe magnet.

Student’s answer:

B USER STUDY MATERIALS

B.1 Pre-test Survey (English translation)

Note that this is an English translation of the German survey questions. In addition to the questions below, the pre-survey began with a consent form, 10-minute video about magnetic fields, 6 multiple-choice physics comprehension questions, and demographic questions for grade level and gender. Contact the authors to request access to these materials.

Generative AI familiarity and usage: The following questions refer to ChatGPT, an artificial intelligence-based chat tool.

- (1) Have you ever heard of ChatGPT, or any other AI-based text generation tool? (Yes / No / Not sure)
- (2) Have you ever used ChatGPT, or any other AI-based text generation tool? (Yes / No / Not sure)
- (3) How many times have you used ChatGPT or similar AI text generation tools in the past year? (Once or twice / About once a month / About once a week / About every day)
- (4) Have you ever used ChatGPT, or any other AI-based text generation tool, for school work (including assignments, exam preparation, concept review, etc.)? (Yes / No)
- (5) How many times have you used ChatGPT or similar AI text generation tools for school work in the past year? (Once or twice / About once a month / About once a week / About every day)
- (6) How do you typically use ChatGPT or similar AI text generation tools for school work? (optional) (600 characters max.)

B.2 Post-test Survey (English translation)

Note that this is an English translation of the German survey questions. In addition to the questions below, the post-survey ended with 13 multiple-choice physics comprehension questions. This paper does not analyze results from the physics comprehension questions. Contact the authors to request access to these materials.

(1) User Experience Questionnaire (UEQ-S)

obstructive	o o o o o o o	supportive
complicated	o o o o o o o	easy
inefficient	o o o o o o o	efficient
clear	o o o o o o o	confusing
boring	o o o o o o o	exciting
not interesting	o o o o o o o	interesting
conventional	o o o o o o o	inventive
usual	o o o o o o o	leading edge

(2) Cognitive load scale

Rate your agreement with the following statements. (Strongly disagree / Somewhat disagree / Neither agree nor disagree / Somewhat agree / Strongly agree)

- (a) The topic covered in the NewtBot activities was very complex.
 - (b) The NewtBot activities covered formulas that I perceived as very complex
 - (c) The NewtBot activities covered concepts and definitions that I perceived as very complex.
 - (d) NewtBot's explanations were very unclear.
 - (e) NewtBot's explanations were very ineffective for my learning process.
 - (f) NewtBot's explanations were full of unclear language.
 - (g) The activity with NewtBot really enhanced my understanding of the topics covered.
 - (h) The activity with NewtBot really enhanced my knowledge and understanding of physics concepts.
 - (i) The activity with Newtbot really enhanced my understanding of the formulas covered.
 - (j) The activity with Newtbot really enhanced my understanding of concepts and definitions.
- (3) NewtBot for performance in school (UTAUT performance expectancy)**

Rate your agreement with the following statements. (Strongly disagree / Somewhat disagree / Neither agree nor disagree / Somewhat agree / Strongly agree)

- (a) Using NewtBot can enable me to accomplish academic tasks more quickly.
- (b) If I use NewtBot, I will increase my chances of getting a better grade in school.
- (c) If I had access to NewtBot, then I would use it to help with my physics school work.

(4) Apprehensions about NewtBot

Do you have any apprehensions about using ChatGPT for school work? Check all that apply.

- No - I do not have any apprehensions.
- Yes - I find it difficult to use the tool effectively.
- Yes - it may violate principles of academic honesty.
- Yes - I may face negative consequences for violating principles of academic honesty.
- Yes - I do not trust that NewtBot's output will be accurate.
- Yes - I do not trust that NewtBot's output will be as nuanced as human-produced output.
- Yes - I fear it will have negative consequences on my learning and understanding of the material.
- Yes - I have concerns about the malicious use of the tool, such as violations of data privacy.
- Yes - I have concerns that the tool will threaten job opportunities for humans.
- Yes - I am uncertain about what its impact could be.
- Yes - I do not feel comfortable taking ownership of output produced by NewtBot.
- Yes - I have another apprehension (please specify) _____